# Fixing the Dark-Energy Equation of State with Two Dimensionless Constants, $\varphi$ and $\pi$: A Falsifiable, Parameter-Frugal Test Against DESI DR2 and Planck 2018

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## Abstract

We present a minimal-parameter framework for the dark sector in which the *shape* of late-time cosmic acceleration is fixed algebraically by two dimensionless constants, the golden ratio $\varphi = (1+\sqrt{5})/2$ and $\pi$, rather than by tuned cosmological parameters. The equation of state follows as bare ratios of structural invariants, $w_0 = -T_r/M_v = -0.839950$ and $w_a = -P_{\rm sc}/I_g = -0.669975$, with no fitted amplitude. The point was fixed in advance and survives *two independent* DESI releases: it sits at $0.05\sigma$ from the DESI DR1 $w_0w_a$CDM posterior and, after the uncertainties tightened, at $0.24\sigma$ from the newer DESI DR2 posterior (Pantheon+; 0.2–$1.8\sigma$ across supernova compilations, $\chi^2_{\rm 2D} = 0.42$)—the prediction did not move while the data improved. Crucially, the two forward predictions that can *refute* the core framework—this fixed $(w_0, w_a)$ point and the tensor ratio $r = \varphi^{-10}$—stand on their own: neither the Hubble resolution nor the growth sector is invoked to state or to kill them. A Markov-chain analysis combining DESI BAO with the global rate of the screening cascade as prior ($H_0^{\rm glob} = 67.962 \pm 0.968$ km s$^{-1}$ Mpc$^{-1}$) yields $H_0 = 67.82 \pm 0.41$ km s$^{-1}$ Mpc$^{-1}$ ($0.68\sigma$ from Planck, $0.34\sigma$ from $H_0^{\rm glob}$). With the background of both models fixed in advance, DESI DR2 BAO gives $\chi^2 = 11.41$ for SSEE and 31.98 for $\Lambda$CDM at its Planck values (13 points); $\Lambda$CDM reaches 10.54 only by lowering $\Omega_m$ to 0.297. The effective-field-theory description is canonical—$\alpha_T = \alpha_M = \alpha_B = 0$ with $\alpha_K = 15.591336$ and a ghost-free sound speed $c_s^2 = 0.021284$—and the inflationary sector predicts $n_s = 1-\varphi^{-7} = 0.965558$ and a tensor-to-scalar ratio $r = \varphi^{-10} = 8.130619 \times 10^{-3}$ that is testable by LiteBIRD and CMB-S4. With its single matter density $\Omega_m = \omega_m/h^2 = 0.308881$, fixed by the algebra, a Boltzmann calculation places the first three acoustic peaks at $\ell = 220, 536, 813$, against $220, 536, 813$ for $\Lambda$CDM. We are explicit about the framework's limits: the absolute vacuum scale enters as a single measured input, the growth amplitude shows no tension with weak lensing once measured on raw data—with $A_s$ fixed to the model's own CMB fit it predicts $S_8 = 0.8273$ against KiDS-Legacy's $0.8265 \pm 0.0176$—and the algebraic origin of the CMB matter density ($\omega_b$, $\omega_c$) is an open problem. Both core forward predictions bear on *unreleased* data: the $(w_0, w_a)$ point is pre-registered to be excluded at $> 3\sigma$ if DESI DR3 shifts it, and $r = \varphi^{-10} = 8.130619 \times 10^{-3}$ is testable by LiteBIRD—either alone can refute the framework. The result is a falsifiable, parameter-frugal description of dark energy whose core predictions rest on $\varphi$ and $\pi$ alone.



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**NOTATION: ALGEBRAIC CONSTANTS USED IN THIS WORK**

All constants below are *dimensionless* and built solely from the golden ratio $\varphi = (1 + \sqrt{5})/2$ and $\pi$. Constants defined in earlier SSEE material but *not used* in this submission (e.g. VITA) are deliberately omitted to keep the register honest.

Each row gives the *symbol* used in the equations, the *name* of the entity in the constant dictionary [1], its *parents* (the entities it is generated from, i.e. its lineage), and the same quantity *reduced to ordinary algebra* in $\varphi$ and $\pi$ alone. Symbols are in one-to-one correspondence with entities: no symbol denotes two entities and no entity carries two symbols. The two columns are not redundant— they are the point. $I_g$ and $K_v$ have *different parents* but reduce to the *same* expression $2(\varphi + \pi)$, which is why they share the value 9.519253 while remaining structurally distinct. The value therefore cannot identify which one an equation uses; only the symbol can.

| Symbol | Name | Parents | Reduced form in $\varphi, \pi$ | Value |
|---|---|---|---|---|
| $\varphi$ | — | seed (UV) | $(1 + \sqrt{5})/2$ | 1.618034 |
| $\pi$ | — | seed (IR) | $\pi$ | 3.141593 |
| $\Omega$ | OMEGA | $\varphi, \pi$ | $\varphi + \pi$ | 4.759627 |
| $\beta$ | BIAL | $\varphi, \pi$ | $(\varphi + \pi)/2$ | 2.379813 |
| KAL | KAL | $\beta, \pi$ | $(\varphi + 3\pi)/2$ | 5.521406 |
| AURA | AURA | $\beta, \varphi$ | $(3\varphi + \pi)/2$ | 3.997847 |
| MIRA | MIRA | AURA | $(3\varphi + \pi)/4$ | 1.998924 |
| $P_{\text{sc}}$ | PYROS | $\Omega, \varphi$ | $2\varphi + \pi$ | 6.377661 |
| $I_g$ | IGNIS | $\pi$, PYROS | $2(\varphi + \pi)$ | 9.519253 |
| $K_v$ | KRYSTOS$_V$ | $\varphi, \pi, \Omega$ | $2(\varphi + \pi)$ | 9.519253 |
| $T_r$ | TRIAL | AURA | $3(3\varphi + \pi)/2$ | 11.993542 |
| $M_v$ | ATLAS | $\varphi, \pi, K_v$ | $3(\varphi + \pi)$ | 14.278880 |

Roles in the equations: $\Omega$ is the stability metric; $\beta$ the base coupling; KAL the structural retention; AURA the screening amplitude; MIRA the mirror ratio entering $f_{\text{screen}}$ (Paper 9); $P_{\text{sc}}$ and $I_g$ the numerator and denominator of $w_a$; $T_r$ and $M_v$ those of $w_0$; and $K_v$ builds $M_v$ but—this is the point of the parent column—*never* appears as a denominator: the denominator of $w_a$ is $I_g$, generated from its own numerator by $I_g = \pi + P_{\text{sc}}$.

Derived background quantities (all dimensionless):

$$w_0 = -\frac{T_r}{M_v} = -0.839950, \qquad w_a = -\frac{P_{\rm sc}}{I_g} = -0.669975, \qquad \Omega_{\rm m}^{\rm dyn} = 1 + w_0 = 0.160050.$$

## I. INTRODUCTION

The standard cosmological model, ΛCDM, fits the available data with six free parameters [2], but pays for this success with a dark sector whose composition and energy scale remain unexplained. The recent DESI DR2 measurements, which favour a dynamical, evolving dark-energy equation of state over a pure cosmological constant [3], sharpen the question: if dark energy is dynamical, what fixes its evolution? A theory that could *predict* the equation of state, rather than fit it, would mark genuine progress.

This paper develops one such proposal. The central hypothesis of the Structural Self-Energy Expansion (SSEE) framework is that the *shape* of the dark sector—the equation-of-state ratios, the effective-field-theory function profile, and the matter density $\omega_m$—is fixed algebraically by two dimensionless constants: the golden ratio $\varphi = (1+\sqrt{5})/2$ and $\pi$. From these, a small set of structural invariants (Table ) is built, and the late-time observables follow as bare ratios with no tuned amplitude.

We are deliberate about what this does and does not claim. SSEE is a *minimal-parameter* framework, not a zero-parameter one: the absolute scale of the vacuum enters as a single measured input (Postulate D, §VII), reducing the count of free CMB inputs from ΛCDM's six to two ($A_s$ and $\tau$; $k = 2$ vs $k = 6$). The dark-sector *shape*, however, carries no free parameter, and this is what makes the framework falsifiable. Once $\varphi$ and $\pi$ are fixed, the equation of state, the spectral index, the tensor-to-scalar ratio, and the EFT slip functions are all determined, and any of them can refute the model.

The paper is organised as a single coherent argument rather than a collection of results. Section II constructs the equation of state from the algebraic invariants and confronts it with DESI DR2, derives the single matter density $\Omega_{\rm m}^{\rm cosm} = \omega_m/h^2 = 0.308881$ and separates it from the equation-of-state number $1 + w_0 = 0.160050$, and reports the Hubble-constant inference. Section IV gives the canonical effective-field-theory description, the inflationary predictions, and the confrontation with Planck 2018, including a Boltzmann cross-check with the model's single matter density ($\omega_m$-direct, $\Omega_{m,\rm CMB} = 0.308881$). Section V treats the growth of structure, including the correction that removed the $S_8$ tension reported by

earlier versions. Section VI sets out the pre-committed prediction register, and Section VII states the model's measured inputs, exact parameter count, and principal open problems. We conclude in Section VIII.

## II. THE ALGEBRAIC CORE: FROM $\varphi, \pi$ TO THE EQUATION OF STATE

### A. Construction of the equation-of-state parameters

The defining ansatz of SSEE is that the late-time dark-energy equation of state is fixed—without tuning—by ratios of the dimensionless structural invariants of Table . Concretely,

$$\begin{aligned} w_0 &= -\frac{T_r}{M_v} = -\frac{3(\varphi+\beta)}{\varphi+\pi+K_v} = -0.839950, \\ w_a &= -\frac{P_{\rm sc}}{I_g} = -\frac{P_{\rm sc}}{\pi+P_{\rm sc}} = -0.66997. \end{aligned} \tag{1}$$

Both quantities are *bare dimensionless numbers*: they are constructed entirely from $\varphi$ and $\pi$ and carry no free normalisation, no fitted amplitude, and no scale input. Under the Chevallier–Polarski–Linder parametrisation $w(a) = w_0 + w_a(1-a)$, Eq. (1) corresponds to an $\alpha$-attractor scalar sector with curvature $\alpha = \varphi^4/3$ (derived in §IV), so the two CPL numbers are not independent fits but two projections of a single rolling field.

### B. Agreement with DESI DR2

We confront Eq. (1) with the four official DESI DR2 $w_0w_a$CDM posteriors [3, eqs. 25–28]. The tightest combination, DESI + CMB + Pantheon+, gives $w_0^{\rm bf} = -0.838 \pm 0.055$, $w_a^{\rm bf} = -0.62^{+0.22}_{-0.19}$ (correlation $\rho = -0.894$, measured directly from the official DESI DR2 chains, Zenodo record 16644577, burn-in 30%, chain weights). Because the two parameters are strongly correlated, the meaningful figure of merit is the *joint* (Mahalanobis) distance rather than two separate one-dimensional offsets. Writing $\mathbf{d} = (w_0 - w_0^{\rm bf},\, w_a - w_a^{\rm bf})$ and the $2\times 2$ covariance $\mathsf{C}$,

$$\chi^2_{\rm 2D} = \mathbf{d}^{\top}\mathsf{C}^{-1}\mathbf{d} = 0.42, \qquad p = 0.81, \qquad \text{equivalent } 0.24\sigma. \tag{2}$$

The component offsets are $\Delta w_0 = -0.04\sigma$ (essentially exact) and $\Delta w_a = -0.26\sigma$. Against the other official combinations (moments and correlations likewise measured from the official

chains: $\rho = -0.905$ to $-0.978$) the joint distance is $1.46\sigma$ (DESY5), $1.76\sigma$ (Union3) and $1.68\sigma$ (no-SN DESI+CMB): the algebraic point is consistent with all four, with the tension range 0.2–$1.8\sigma$ set by the supernova compilation. The result is robust to the correlation ($\rho \in [-0.92, -0.70]$ would shift the Pantheon+ figure by $< 0.2\sigma$; the measured value is used). We stress that this agreement is obtained with *zero* hand-tuned dark-energy parameters: the only inputs are $\varphi$ and $\pi$. The fixed $\Lambda$CDM point $(-1, 0)$, by contrast, sits at 2.6–$3.9\sigma$ from every official compilation (Table I): SSEE is closer to the data than $\Lambda$CDM for all four supernova compilations—a like-for-like comparison of two zero-parameter predictions against the same measurement, neither of which tunes $(w_0, w_a)$.

TABLE I. Mahalanobis distance of the fixed SSEE point $(-0.840, -0.670)$ and the fixed $\Lambda$CDM point $(-1, 0)$ from each official DESI DR2 $w_0w_a$CDM best-fit ($2\times2$ covariance, correlation measured directly from the public chains, Zenodo 16644577). Neither point is fitted; both are predictions. SSEE is closer to the data than $\Lambda$CDM for all four supernova compilations.

| DESI DR2 combination | $\rho$ | SSEE distance | $\Lambda$CDM distance |
|---|---|---|---|
| DESI+CMB+Pantheon+ | $-0.894$ | $\mathbf{0.24}\sigma$ | $2.58\sigma$ |
| DESI+CMB+DES Y5 | $-0.905$ | $\mathbf{1.46}\sigma$ | $3.93\sigma$ |
| DESI+CMB+Union3 | $-0.933$ | $\mathbf{1.76}\sigma$ | $3.38\sigma$ |
| DESI+CMB (no SN) | $-0.978$ | $\mathbf{1.68}\sigma$ | $2.62\sigma$ |

*a. Honest caveat: this is agreement with BAO+SN, not with CMB-only.* The $0.24\sigma$ figure is specific to the DESI DR2 dynamical-dark-energy posterior, which is dominated by low-redshift data ($z \simeq 0.3$–2.3), where dark energy actually drives the expansion, and the quoted range 0.2–$1.8\sigma$ reflects the known spread among supernova compilations. Against a *Planck CMB-only* $w_0w_a$CDM analysis, which constrains the late-time equation of state only through its extrapolation from $z \simeq 1100$, the same $(w_0, w_a)$ is disfavoured. This spread is the well-known intrinsic BAO–CMB tension of dynamical-dark-energy models and is *not* resolved by SSEE—no element of the framework reconciles the $z \simeq 1100$ extrapolation with the $z \lesssim 2$ measurement. We report the DESI agreement as what it is: a clean, parameter-free match to the dataset that directly probes the dark-energy-dominated epoch.

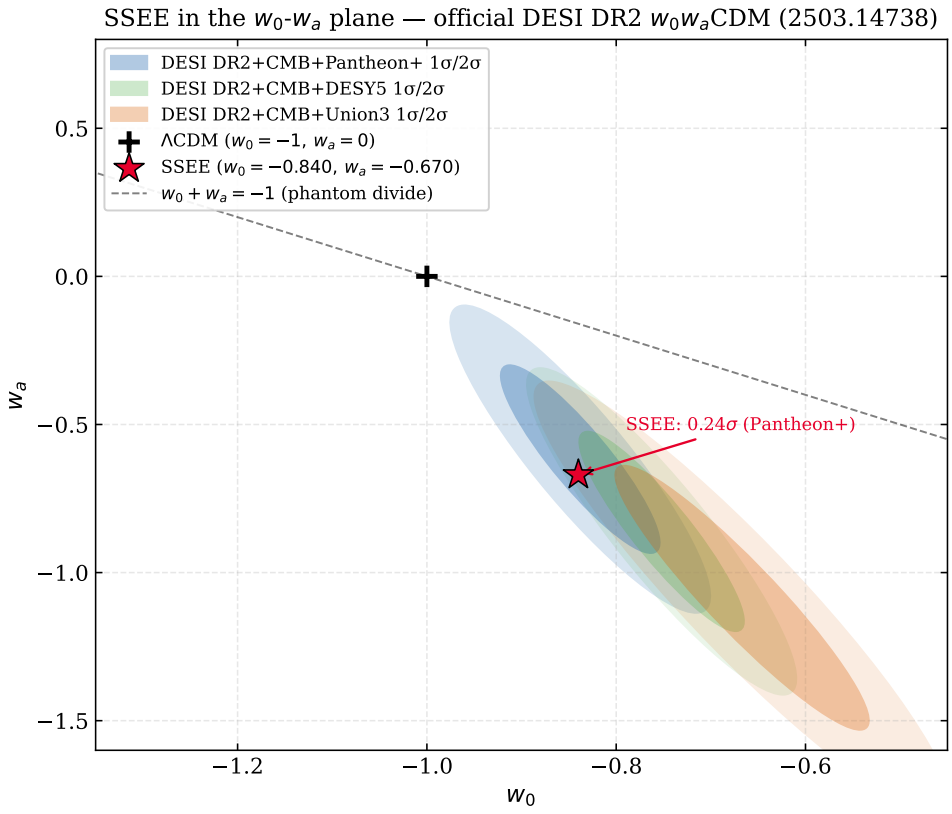



FIG. 1. The SSEE prediction in the $w_0$–$w_a$ plane. The single point $(w_0, w_a) = (-T_r/M_v,\, -P_{\rm sc}/I_g) = (-0.840, -0.670)$ is fixed entirely by $\varphi$ and $\pi$, with no dark-energy parameters adjusted to the data. The shaded contours are the official DESI DR2 $w_0w_a$CDM posteriors (Pantheon+, DESY5, Union3). The SSEE point falls inside the Pantheon+ $1\sigma$ region; the joint Mahalanobis distance is $\chi^2_{\rm 2D} = 0.42$, an equivalent $0.24\sigma$ agreement (Eq. 2), and within 1.5–$1.8\sigma$ of the other compilations.

### C. A look-elsewhere test: how special are these two ratios?

A natural objection to any algebraic coincidence is the look-elsewhere effect: if one is free to combine $\varphi$ and $\pi$ in arbitrarily many ways, hitting two target numbers is unremarkable. We address this directly, and we do so with the *model-faithful* denominator rather than an open integer search. SSEE does not draw $w_0$ and $w_a$ from an unbounded space of expressions; it draws them from a *closed dictionary* of 55 named structural invariants—the *complete* Genesis set, not a hand-picked subset—specified as a fixed set rather than

expanded to accommodate the data.[1] The honest question is therefore: among *all* ratios $A/B$ of named invariants in this fixed dictionary, how many land on $|w_0| = 0.839950$ and on $|w_a| = 0.669975$?

Enumerating every ordered pair over the complete dictionary yields 490 distinct ratios in the interval $(0, 5]$. We then count how many fall within a tolerance of each target. Crucially, the SSEE assignments are not near-misses but *exact* algebraic identities—$|w_0| = \text{AURA}/\Omega = T_r/M_v$ and $|w_a| = \text{PYROS/IGNIS} = P_{\text{sc}}/(\pi + P_{\text{sc}})$, both to machine precision ($|d| = 0$)—so the physically relevant tolerance is that of the exact identity ($\pm 0.0005$, $|d| = 0$), not an arbitrary band. At that precision each target is unique:

$$|w_0| = 0.839950 : \ \text{1 of 490}, \qquad |w_a| = 0.669975 : \ \text{1 of 490} \quad (\pm 0.0005). \tag{3}$$

Table II reports the full tolerance sensitivity: uniqueness holds in the strict, sub-percent regime the *exact* match occupies, and relaxing the band above DESI precision predictably densifies accidental hits. At measurement precision each equation-of-state parameter is *essentially unique*—the opposite of a dense coincidence forest, a stronger statement than a generic $(a\varphi + b\pi)/(c\varphi + d\pi)$ scan, which by construction inflates the denominator with expressions the model never uses.

TABLE II. Look-elsewhere sensitivity to the matching tolerance, computed over the *complete* 55-constant Genesis dictionary (490 distinct ratios in $(0, 5]$; constants sharing a value collapse to 25 distinct numbers). The DESI assignments are exact identities ($|d| = 0$); the relevant band is $\pm 0.0005$ (the exact-identity precision), where each parameter is unique. The count is robust to the planned future dimensional copies (QUINTAL–DECAL): they add no hit within $\pm 0.0005$ of either target.

| Tolerance | hits on $\lvert w_0\rvert = 0.839950$ | hits on $\lvert w_a\rvert = 0.669975$ |
|---|---|---|
| $\pm 0.0005$ | 1 | 1 |
| $\pm 0.001$ | 1 | 2 |
| $\pm 0.002$ | 1 | 4 |
| $\pm 0.005$ | 2 | 5 |
| $\pm 0.01$ | 4 | 9 |

[1] The dictionary is the full Genesis constant set [1], counted over *every* invariant with none pruned—conservative, since it can only inflate the accidental-hit count. The 55 names collapse to 25 distinct numbers (e.g. IGNIS and KRYSTOS coincide, SOLAR = MAR), so the count deduplicates by value.

The two ratios are moreover not independent draws. Each is an intra-lineage ratio of the closed dictionary: $|w_0|$ = AURA/$\Omega$ is the copy-branch root over the scaffold, and $|w_a|$ = PYROS/IGNIS has a denominator generated directly from its own numerator by the no-self-sum law (IGNIS = $\pi$ + PYROS). The two CPL numbers are two projections of one rolling field (§II A), not two separately fitted constants, so the joint look-elsewhere is governed by a single shared structure rather than by two free choices. We report this as evidence of internal economy, not as a claim that the dictionary itself is derived from first principles—that derivation (the dynamical origin of the two lineage laws) remains the open mechanism tracked in §VII.

*a. The single near-value, and why it is not a rival.* Honesty requires naming the one entry that turns the $|w_a|$ count from 1 to 2 as the band is loosened from $\pm 0.0005$ to $\pm 0.001$ (Table II): it is SOLAR/NYX = 0.66933, which sits $|d| = 6.4 \times 10^{-4}$ from $|w_a|$ = PYROS/IGNIS = 0.669975. Two independent facts demote it from rival to shadow. *First, construction:* the canonical ratio PYROS/IGNIS is intra-lineage—the denominator is generated directly from the numerator, IGNIS = $\pi$ + PYROS, so numerator and denominator lie on the same $\pi$-branch chain—whereas SOLAR and NYX = $(\varphi + 7\pi)/2$ (the retention-branch floor, charge $q = -3$) share no generative relation; their ratio is a numerical coincidence across lineages, not a structural identity. The selection is therefore made by lineage, not by tolerance. *Second, transcendence:* the proximity is the shadow of the numerical near-coincidence $2\varphi \approx \pi$ ($2\varphi = 3.236$, $\pi = 3.142$), the exact residual scaling as $(2\varphi - \pi)(\varphi - \pi) \neq 0$. Because $\pi$ is transcendental, $2\varphi - \pi$ is irrational and non-zero for all time, so this near-value is *forbidden* from ever collapsing onto an exact identity—by the very property of $\pi$ that grounds the no-self-sum law of Appendix A. The law that generates NYX is thus the same law that guarantees it can never become a genuine second identity: the near-miss is structural, not accidental, and disclosing it strengthens rather than weakens the count.

*b. Scope: this test is for the core, not the extension.* The look-elsewhere statement above applies to the dark-energy core $(w_0, w_a)$, where the dictionary grammar is fixed by construction rules (no-self-sum plus copy-law) that precede and are independent of the DESI targets. No $\varphi$-DM mass multiplier is left to test: that construction is retracted. Its warrant had been *falsifiability* — a pre-registered free-streaming scale that DESI Y3 and Euclid would confirm or exclude. That warrant was insufficient, and the episode is instructive:

falsifiability is the minimum requirement for a hypothesis to be scientific, not evidence that the entity it concerns exists.

### D. The cosmic matter fraction: one matter density

The equation of state (1) fixes the combination that earlier versions called the *dynamical* matter fraction,

$$\Omega_{\rm m}^{\rm dyn} \;=\; 1 + w_0 \;=\; 0.160050, \tag{4}$$

a number from the equation of state, not a density. It is much lower than the matter density inferred from the acoustic peaks, and it is not meant to match it. Under the $\omega_m$-direct reframe (retiring the former matter-bridge open problem, OP-8; §VII) the CMB matter fraction is the standard physical observable, built piece-by-piece from SSEE algebra with *no* bridging factor,

$$\Omega_{\rm m}^{\rm cosm} \;=\; \frac{\omega_m}{h^2} \;=\; \frac{\omega_b + \omega_c + \omega_\nu}{h^2} \;=\; \frac{0.14267}{(67.96214/100)^2} \;=\; 0.308881, \qquad \omega_c = \mathrm{KAL}_0\,\omega_b\,n_s, \tag{5}$$

the matter fraction that enters the background expansion $E(z)$, the Poisson equation, and the CMB — the model's only matter density. The equation-of-state number $1 + w_0 = 0.160050$ (tested by DESI through $w_0$) and $\omega_m$ (a density, tested by the CMB) are two independent algebraic numbers of different kinds, no longer tied by a factor.

*a. No bridging factor: the CMB density is built from algebra.* Each piece of $\omega_m$ is an SSEE algebraic value with no fit: the baryon density $\omega_b = (\pi - \varphi)/[3(\varphi + \pi)^2] = 0.022418$ and the cold-matter density $\omega_c = \mathrm{KAL}_0\,\omega_b\,n_s = 0.119514$ (a forward identity, already in Paper 1; $0.41\sigma$ from Planck). The neutrino term completes the total, $\omega_\nu = \Sigma m_\nu^{\rm active}/C_\nu = 0.06849/93.14 = 0.000735$, using the same active-neutrino sum $\Sigma m_\nu^{\rm active} = 0.06849$ eV and the same relic closure constant $C_\nu = 93.14$ eV (PDG, $N_{\rm eff} = 3.046$),[2] so that

$$\omega_m = \omega_b + \omega_c + \omega_\nu = 0.14267, \qquad \Omega_{\rm m}^{\rm cosm} = \frac{\omega_m}{h^2} = 0.308881 \quad (h = H_0^{\rm glob}/100 = 0.67962). \tag{6}$$

The neutrino piece is not optional: dropping it returns 0.30729, a 0.6% shift, so a reader who reproduces the number must include all three terms (a three-second check ships in

[2] $\Sigma m_\nu^{\rm active}$ is not a fitted input: it follows from the algebraic chain $\Sigma m_\nu^{\rm active} = R_2\,\omega_b\,C_\nu/(\tau_\Pi H_0)$ with $R_2 = \Omega/(\mathrm{KAL}_0 T_r)$ (Paper 6). No independent neutrino parameter is introduced.

`ssee_core.py`). The earlier presentation tied $\Omega_{\rm m}^{\rm cosm}$ to $\Omega_{\rm m}^{\rm dyn}$ through a factor-of-two bridge MIRA = AURA/2 = 1.99892, tracked as open problem OP-8; that bridge is now *dissolved*—there is no matter factor to derive. The residual honest open question is only the UV origin of the $\omega_c$ identity. MIRA survives elsewhere, in the late-time Hubble screening fraction $f_{\rm scr} = s_K/(3\,\mathrm{MIRA})$ (Paper 9), where it is the "Mirror Ratio".

*b. One matter density.* The equation-of-state number $1+w_0 = 0.160050$ and the matter density $\Omega_{\rm m}^{\rm cosm} = \omega_m/h^2 = 0.308881$ are two independent algebraic predictions, not two free numbers and not two matter densities: $1 + w_0$ never enters a density slot. Earlier versions of this paper put it there in a Boltzmann run and reported that the acoustic peaks then degrade; that comparison tested a density the model does not contain, and it is withdrawn.

Figure 2 makes the same point on the drag-epoch sound horizon itself: the physical CMB drag scale set by the full algebraic $\omega_m$ is 147.17 Mpc at $\Omega_{\rm m}^{\rm cosm} = 0.308881$ ($\omega_m$-direct), consistent with the Planck-inferred $147.09 \pm 0.26$ Mpc ($0.32\sigma$), and matching ΛCDM at the Planck 2018 parameters (147.10 Mpc, same code). The equation-of-state number $1 + w_0 = 0.160050$ does not enter the geometry.

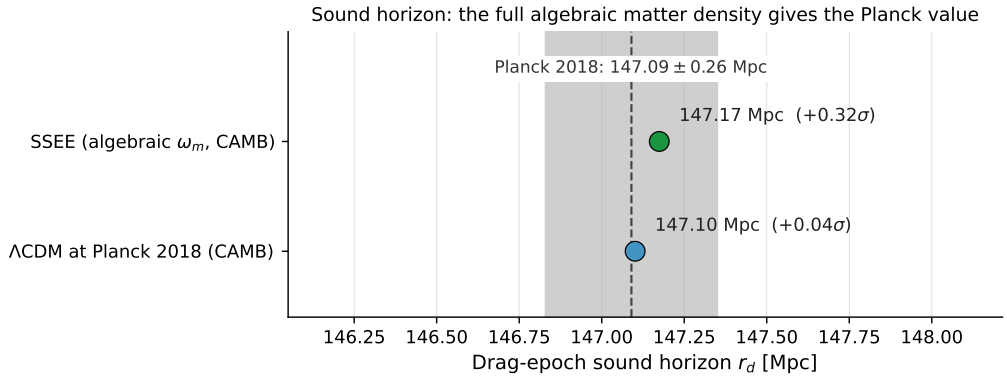



FIG. 2. The physical sound horizon. Set by the full algebraic $\omega_m$ (green, 147.17 Mpc at $\Omega_{\rm m}^{\rm cosm} = 0.308881$, $\omega_m$-direct), in agreement with Planck 2018 ($147.09 \pm 0.26$ Mpc, grey band) and with ΛCDM at the Planck 2018 parameters (blue, 147.10 Mpc, CAMB). This is the visual counterpart of the Boltzmann test above: the full matter density that restores the acoustic peaks also restores the BAO standard ruler.

## III. BAYESIAN VALIDATION: MCMC AGAINST DESI DR2

The algebraic predictions of §II are confronted with data in a full Markov-chain Monte Carlo analysis. The dark-energy sector contributes no sampled parameters—$w_0$, $w_a$ and the matter density $\Omega_{\rm m}^{\rm cosm}$ are fixed by Eqs. (1)–(5)—so the chain effectively samples $H_0$ (and $\Omega_b h^2$) against the DESI DR2 BAO data with a CMB-derived prior, using the affine-invariant ensemble sampler [4]. We ran 100 walkers $\times$ 25,000 steps (seed 42).

*a. $H_0$ as one quantity in two stages.* We first anchor the early-time geometry at the CMB-optimal value

$$\begin{aligned} H_0^{\rm glob} &= H_{\rm SH0ES}(1-f_{\rm screen}) = 67.962 \pm 0.968 \ {\rm km\,s^{-1}\,Mpc^{-1}}, \\ \frac{H_0^{\rm glob}}{\rm km\,s^{-1}\,Mpc^{-1}} &\simeq 3(\varphi+\pi)^2, \end{aligned} \tag{7}$$

which (with $\omega_b, \omega_c$ fixed by SSEE algebra) lies $0.90\sigma$ from the $H_0$ that minimises the Planck `plik_lite` likelihood for the SSEE background ($67.87 \pm 0.10$, profiled over $A_s, \tau$); at this anchor the sound horizon and acoustic scale are both well within tolerance, $r_d = 147.17$ Mpc ($0.32\sigma$) and $\theta_* = 0.59667^\circ$ ($1.00\sigma$). Letting the MCMC then correct $H_0$ against the low-redshift BAO (corrected DESI DR2 vector, block-diagonal covariance with the official $r_{M,H}$ correlations) yields the posterior

$$H_0^{\rm post} = 67.82 \pm 0.41 \ {\rm km\,s^{-1}\,Mpc^{-1}}, \qquad \Omega_b h^2 = 0.02198 \pm 0.00048, \tag{8}$$

with $r_d = 147.17$ Mpc ($0.32\sigma$, $\omega_m$-set, $H_0$-invariant) at the posterior value. These are two stages of the same quantity—a CMB anchor and a BAO-corrected posterior—and with the total matter density in the background geometry they *coincide*. (The earlier MIRA-anchored 67.037/66.533 stages, the 67.159 posterior from the mislabelled DR1 vector, and the 66.412 obtained with the cold sector 0.160 in the geometry, are all superseded.)

*b. Status of the algebraic anchor.* We are explicit about how a dimensionful value is obtained from Eq. (7), and about which observable is the input. On its own, $3(\varphi+\pi)^2 = 67.962$ is a pure number with no units; deriving the absolute scale of $H_0$ from first principles is an open problem for every cosmological model, not only ours, and we make no claim to have solved it. We therefore do not assert the pure number as physical and then show that it explains the data. Instead we start from the measured SH0ES *local* rate, $H_0^{\rm SH0ES} = 73.04 \pm 1.04 \ {\rm km\,s^{-1}\,Mpc^{-1}}$ [5], and de-screen it with the algebraically-derived (not fit) fraction

$f_{\rm scr}^{\rm IR} = s_K^{\rm IR}/(3\,\mathrm{MIRA}) = (\pi-\varphi)/\Omega_{\rm m}^2 = 0.067253$ (Paper 9): $H_0^{\rm glob,IR} = H_0^{\rm SH0ES}(1-f_{\rm scr}^{\rm IR}) = 68.13\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$, within $0.17\sigma$ of the pure number 67.962—the model-independent floor, using only the IR screening term and no free parameter. Paper 10 sharpens this: with the UV-completed fraction $f_{\rm scr}^{\rm UV} = s_K^{\rm full}/(3\,\mathrm{MIRA}) = 0.069522$ (conditional on the derivation of $M^4 = 5\varphi^8\rho_{\rm crit}$ from UV–IR $\varphi/\pi$ separability, Postulate C.1), the identical inversion gives $H_0^{\rm SH0ES}(1-f_{\rm scr}^{\rm UV}) = 73.04\times(1-0.069522) = 67.96214\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$, reproducing the pure-number target $3(\varphi+\pi)^2 = 67.9621$ to four decimals. We quote the comparison at four decimals and not further because the input is quoted at two: $H_0^{\rm SH0ES} = 73.04$. We do not quote the reverse operation $3(\varphi+\pi)^2/(1-f_{\rm scr}^{\rm UV})$ as a prediction, because $3(\varphi+\pi)^2$ is a pure number and carries no units, so it cannot serve as the input of a dimensionful cascade. The only directly measured expansion rate is the local one, and Planck's is inferred within ΛCDM rather than measured; SH0ES is therefore the input throughout. The leftover, $4\times10^{-6}\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$, lies below the resolution of the datum it is compared against; whether a future, more precisely quoted local measurement drives it down or opens it up is an empirical question, not one we can settle here. We say "to the quoted precision" rather than "exactly" for that reason. We are deliberately precise about what is independent of what. The de-screening is *external*: it consumes only the measured SH0ES rate and a screening fraction derived beforehand, with no SSEE value of $H_0$ as input—but its two variants are one datum under two prescriptions, not two independent measurements. The CMB/BAO chain, which never touches SH0ES, adopts the anchor as its prior, so its posterior is a *consistency* test rather than a third confirmation: its content is that DESI DR2 does not pull the value away ($\chi^2_{\rm BAO} = 10.43$ over 13 points at the posterior, 11.41 at the anchor, against $\simeq 726$ for the incorrect background geometry), leaving a residual displacement of $0.34\sigma$. Replacing the anchor prior with a Planck one removes even that dependence: the data then pull $H_0$ *upward* from 67.36 to $67.62\pm0.35$, towards the anchor rather than away from it. That pattern—one external datum and two independent-prior checks that decline to move the value—is the anchor's physical content, not a claim that $3(\varphi+\pi)^2$ literally equals a dimensionful rate. In Planck units $H_0 \sim 10^{-61}$ and no such coincidence appears; the dimensional origin—the bridge from the megaparsec scale to fundamental units—remains open (§VII). The anchor is falsifiable on both fronts: a CMB/BAO posterior displaced from it by more than $2\sigma$, or a SH0ES-derived local rate that moves the unconditional IR floor outside $\sim 1\sigma$, would undermine it.

*c. Anchor and posterior agree.* With the total gravitating matter density $\Omega_{\rm m}^{\rm cosm} = 0.308881$ in $E(z)$ and $r_d$, the BAO-corrected posterior (67.82) and the CMB anchor $H_0^{\rm glob}$ (67.962) agree to $0.34\sigma$: DESI DR2 is *consistent with* the global rate rather than pulling against it. All early-time observables are simultaneously healthy—$r_d = 147.17$ Mpc ($0.32\sigma$) and, evaluating the acoustic scale at the posterior, $100\,\theta_* = 1.04099$ ($0.34\sigma$). The apparent $2.9\sigma$ "BAO–CMB split" and the $13.9\sigma$ $\theta_*$ figure reported in earlier drafts were the footprint of a single error—placing the cold dynamical sector $\Omega_{\rm m}^{\rm dyn} = 0.160050$ in the background geometry, where the total density belongs. With that corrected, there is no split to report: the late-time BAO and the early-time acoustic geometry point to the same $H_0$.

*d. Model comparison against the natural competitors.* Because the dark-energy sector carries no sampled parameters, the honest comparison is not against ΛCDM alone but against the whole family a referee would reach for. On the identical DESI DR2 vector, the information criteria are: SSEE is favoured over both ΛCDM ($\Delta$BIC $= 7.25$) and CPL (6.14)

TABLE III. Bayesian model comparison on DESI DR2 (100 walkers × 25,000 steps). Δ values are relative to SSEE; positive favours SSEE.

| Model | $k$ | $\ln P_{\rm MAP}$ | BIC | ΔBIC | ΔAIC |
|---|---|---|---|---|---|
| SSEE | 2 | −5.52 | 16.58 | 0.00 | 0.00 |
| ΛCDM | 3 | −7.76 | 23.83 | +7.25 | +6.48 |
| CPL | 5 | −4.43 | 22.72 | +6.14 | +3.82 |

by parsimony, with $\Delta$DIC $= -6.47$ and a Savage–Dickey factor $\ln B = 2.34$ (substantial on the Jeffreys scale); a leave-one-out cross-validation at $z \geq 1$ ranks SSEE ahead of ΛCDM ($\chi^2_r = 0.26$ versus 0.71; CPL's training fit runs to the edge of the $H_0$ prior and is not quoted). The preference is driven by fewer parameters at equal fit, not by a superior $\chi^2$: the competitor that *can* bend to the data (CPL, $k = 5$) is penalised for the freedom SSEE does not use.

*e. Each probe with the background fixed in advance.* The comparison above lets ΛCDM fit its background to DESI. Table IV instead fixes the background of both models before looking at each probe—SSEE at its algebraic point, ΛCDM at the Planck 2018 best fit—and reads the $\chi^2$ of each probe with no cosmological parameter fitted. One probe separates the two models by more than $\Delta\chi^2 = 9$: DESI DR2 BAO, with $\chi^2 = 11.41$ for SSEE and

31.98 for ΛCDM over 13 points ($\Delta\chi^2 = -20.6$). When ΛCDM is allowed to fit DESI, it reaches $\chi^2 = 10.54$ for 11 degrees of freedom by lowering $\Omega_m$ from 0.3152 to $0.297 \pm 0.009$; SSEE reaches 11.41 with nothing fitted. The other probes do not separate the models. The supernova compilations share objects and the two CMB-lensing probes share sky, so the rows are not summed.

TABLE IV. $\chi^2$ of SSEE and ΛCDM probe by probe, read from the analysis logs. Block A: background fixed in both models in advance (ΛCDM at the Planck 2018 best fit). Block B: ΛCDM fits cosmological parameters that SSEE does not have (column "fitted", SSEE / ΛCDM). $\Delta\chi^2 = \chi^2_{\rm SSEE} - \chi^2_{\Lambda\rm CDM}$. "$N$" is the number of degrees of freedom where the log records it. [a]Seven nuisance parameters minimised identically in both models. [b]TATT intrinsic alignments (the survey's baseline); with NLA the difference is $\Delta\chi^2 = 0.41$.

| Probe | Type | $N$ | fitted | $\chi^2_{\rm SSEE}$ | $\chi^2_{\Lambda\rm CDM}$ | $\Delta\chi^2$ |
|---|---|---|---|---|---|---|
| *A. Background fixed in both models* | | | | | | |
| Pantheon+ | SN | 1579 | 0 / 0 | 1386.62 | 1387.33 | −0.70 |
| DES-SN5YR | SN | 1819 | 0 / 0 | 1629.44 | 1632.41 | −2.97 |
| Union3 | SN | 21 | 0 / 0 | 23.85 | 26.37 | −2.52 |
| Union3.1 | SN | 21 | 0 / 0 | 26.60 | 28.83 | −2.24 |
| ACT DR6 | CMB lensing | 10 | 0 / 0 | 14.01 | 14.20 | −0.20 |
| SPT-3G D1 KK[a] | CMB lensing | 10 | 0 / 0 | 17.76 | 18.31 | −0.54 |
| DES Y3[b] | 3×2pt | – | 0 / 0 | 516.11 | 517.29 | −1.18 |
| DESI DR2 BAO | BAO | 13 | 0 / 0 | 11.41 | 31.98 | −20.58 |
| *B. ΛCDM fits parameters that SSEE does not have* | | | | | | |
| DESI DR2 BAO | BAO | 13 / 11 | 0 / 2 | 11.41 | 10.54 | 0.87 |
| Planck plik_lite | CMB | – | 2 / 6 | 1003.59 | 1003.60 | −0.01 |
| Planck plik (full) | CMB | – | 2 / 6 | 2768.45 | 2771.23 | −2.78 |
| KiDS-Legacy | shear | – | 0 / 1 | 417.97 | 418.95 | −0.98 |

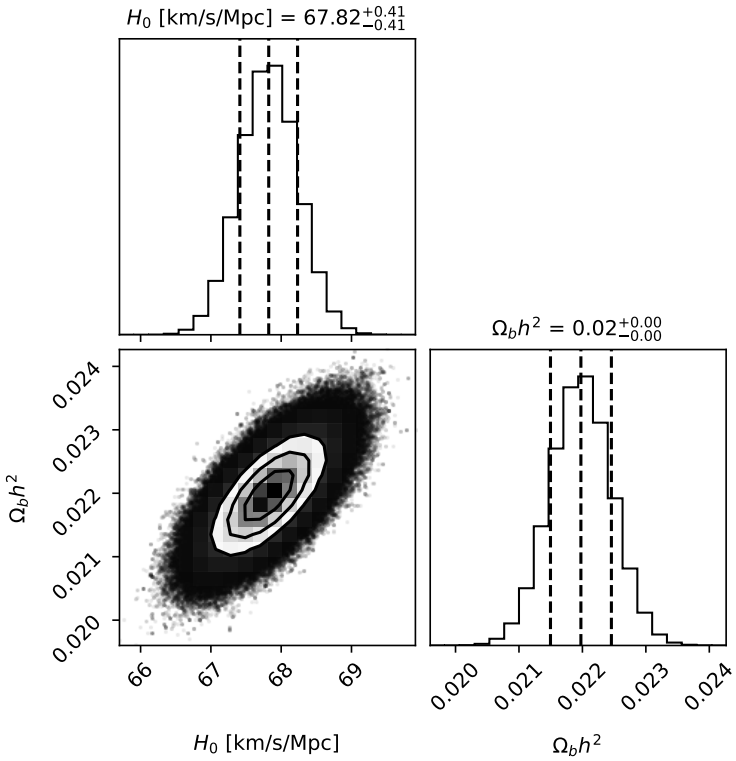



FIG. 3. Marginalised posterior from the MCMC analysis (100 walkers $\times$ 25,000 steps). With the entire dark-energy sector fixed by $\varphi$ and $\pi$, the chain samples only $H_0$ and $\Omega_b h^2$ against DESI DR2 BAO with $H_0^{\rm glob}$ as prior. The posterior is $H_0 = 67.82 \pm 0.41$ km s$^{-1}$ Mpc$^{-1}$ and $\Omega_b h^2 = 0.02198 \pm 0.00048$ (Eq. 8), agreeing with the anchor to $0.34\sigma$. Dashed lines mark the prior centres ($H_0^{\rm glob} = 67.962$ km s$^{-1}$ Mpc$^{-1}$ and the BBN $\Omega_b h^2 = 0.02218$ [6]); the 16/50/84% quantiles are indicated.

## IV. THE CMB AND THE EFFECTIVE FIELD THEORY

The equation-of-state construction of §II fixes the late-time expansion history. To test the framework against the cosmic microwave background and large-scale structure we must specify its behaviour at the level of linear perturbations. We do so through an effective field theory (EFT) of dark energy whose functions are again fixed by $\varphi$ and $\pi$, with no perturbation-level parameters introduced beyond those already present in the background.

### A. The effective field theory of the dark sector

In the EFT-of-dark-energy language [7–9] the linear dynamics of a single scalar degree of freedom are controlled by four functions $\alpha_{\rm K}$ (kineticity), $\alpha_{\rm B}$ (braiding), $\alpha_{\rm M}$ (Planck-mass run) and $\alpha_{\rm T}$ (tensor speed excess). For the operative Lagrangian of Paper 7, $K(X) = c_1 X + c_2 X^2$, minimally coupled and with no potential, they are

$$\alpha_{\rm T} = \alpha_{\rm M} = \alpha_{\rm B} = 0, \qquad \alpha_{\rm K} = 15.591336, \tag{9}$$

so that gravitational waves propagate at the speed of light ($\alpha_{\rm T} = 0$, consistent with GW170817 [10]) and the only non-trivial function is the kineticity, evaluated with the dark-energy *density* $\Omega_{\rm DE} = 1 - \Omega_m = 0.691119$.

A distinct quantity, often confused with it, enters the Hubble screening of Paper 9:

$$s_K \equiv -\frac{1}{\rho_\phi}\frac{dp_\phi}{dN} = -3w_0(1+w_0) = 3\, s_{\rm DE}\, s_m = 0.403302, \tag{10}$$

the fractional dilution rate of the dark-energy *pressure* per e-fold. Here $s_{\rm DE} = |w_0| = 0.839950$ and $s_m = 1 + w_0 = 0.160050$ are saturations — numbers of the equation of state, carrying no $H$ — not densities. Because $s_K$ is pure equation of state, the screening amplitude $f_{\rm scr} = s_K/(3\,\mathrm{MIRA})$ is invariant under changes of the expansion anchor, whereas a kineticity would not be. The Lagrangian above contains no coupling of the scalar to matter. What remains open is tracked as OP-24: the same action fixes $w_0$ but does not produce $w_a = -0.669975$.

### B. Inflationary sector: $\alpha$-attractor

The primordial perturbations are described by an $\alpha$-attractor [11] with curvature

$$\alpha = \frac{\varphi^4}{3} = 2.284701. \tag{11}$$

For the attractor universality class, $n_s = 1 - 2/N_*$ and $r = 12\alpha/N_*^2$. We take the number of $e$-folds to be $N_* = 2\varphi^7 = 58.068884$. The coefficient 2 is not a free choice. Within the observationally allowed window $[50, 60]$ the products $m\varphi^n$ with small integer $m$ are $2\varphi^7$, $3\varphi^6$, $5\varphi^5$ and $8\varphi^4$; of these, $m = 2$ is the only one for which the spectral index comes out as a pure power of $\varphi$, precisely because that 2 cancels against the 2 in $n_s = 1 - 2/N_*$ (the others give

| Spectrum | SSEE $\chi^2_r$ | $\Lambda$CDM $\chi^2_r$ | $N$ |
|---|---|---|---|
| TT | 1.042 | 1.043 | 1971 |
| TE | 1.040 | 1.040 | 1967 |
| EE | 1.040 | 1.039 | 1967 |
| $\phi\phi$ (lensing) | 0.720 | 0.757 | 9 |

TABLE V. CMB confrontation against Planck 2018. SSEE reproduces the acoustic spectra at a quality indistinguishable from $\Lambda$CDM, with acoustic peaks at $\ell = 220, 536, 813$. The combined $\Delta\mathrm{BIC}(\mathrm{SSEE} - \Lambda\mathrm{CDM}) = -35.0$ over $N = 5914$ data points (diagonal, $\omega_m$-direct) reflects the two-parameter economy of the background.

$1-\varphi^{-6.84}$, $1-\varphi^{-6.90}$ and $1-\varphi^{-6.88}$). The same coefficient simultaneously renders $r$ a pure power. $N_* = 2\varphi^7$ is therefore the unique member of that family for which *both* observables reduce to pure powers of $\varphi$, and the two resulting identities are exact:

$$n_s = 1 - \varphi^{-7} = 0.965558, \qquad r = \varphi^{-10} = 8.130619 \times 10^{-3}. \tag{12}$$

The spectral index sits squarely on the Planck 2018 measurement ($n_s = 0.9649 \pm 0.0042$; 2); the tensor-to-scalar ratio $r = \varphi^{-10}$ is a pre-committed, falsifiable prediction (§VI) well below the current bound $r < 0.036$ and within reach of LiteBIRD and CMB-S4. We note that, within the framework, $n_s$ and $r$ are postulated of this form and $\alpha = \varphi^4/3$ then follows as an exact algebraic consequence; we do not claim an independent derivation of the exponent 7 from the potential (see §VII).

### C. Confrontation with Planck 2018

With the background of §II and the EFT functions of Eq. (9), we computed the CMB angular power spectra with the `CLASS` Boltzmann code [12] and compared against Planck 2018 [13] (TT, TE, EE and lensing). Table V reports the reduced $\chi^2$.

*a. Which likelihood the model comparison rests on.* We are explicit about this because the two numbers differ and only one of them is a model comparison. The $\chi^2_r$ of Table V and the associated $\Delta\mathrm{BIC} = -35.0$ use a *diagonal* covariance. That is adequate for displaying spectral agreement, but not for selecting between models: neglecting the Planck bandpower

correlations overstates the number of independent data points, and since the BIC penalty enters as $k \ln N$, it inflates precisely the term that rewards the smaller $k$. We therefore also evaluate the compressed Planck likelihood `plik_lite` (TT+TE+EE, 613 bandpowers, nuisance parameters marginalised by the Planck Collaboration) together with the low-$\ell$ TT and EE likelihoods, for $N = 669$ data points, which gives $\chi^2_{\rm SSEE} = 1003.586$ against $\chi^2_{\Lambda\rm CDM} = 1003.596$ (with its own $\sum m_\nu = 0.06$ eV) with $\{A_s, \tau\}$ minimized in both—an equal fit ($\Delta\chi^2 = -0.010$, statistically zero) for four fewer free parameters—and

$$\Delta\mathrm{BIC}(\mathrm{SSEE} - \Lambda\mathrm{CDM}) = -26.03 \qquad (\texttt{plik_lite}, k = 2 \text{ vs } 6). \tag{13}$$

The diagonal $-35.0$ is retained for continuity with Paper 3 and should be read as illustrative. One residual dependence would remain—`plik_lite` marginalises its nuisance parameters within a ΛCDM analysis—so we also evaluated the *full* `plik` likelihood (TTTEEE + lowl TT + lowl EE + lensing, $N = 2354$) with the ∼20 nuisance parameters free, and take the *minimum* $\chi^2$ of each model:

$$\chi^2_{\rm SSEE} = 2768.4, \quad \chi^2_{\Lambda\rm CDM} = 2771.2, \quad \Delta\mathrm{BIC} = -33.83 \qquad (\texttt{plik full}, k = 2 \text{ vs } 6). \tag{14}$$

The two fits are indistinguishable ($\Delta\chi^2 = -2.78$); the gain is the parameter count. Counting conservatively $k = 4$ for SSEE gives $-18.31$. As a validation, a separate chain that also samples $H_0$ returns $H_0 = 67.88 \pm 0.10$ km s$^{-1}$ Mpc$^{-1}$, $0.81\sigma$ from $H_0^{\rm glob} = 67.962$: fixing $H_0$ costs no fit quality.

*b. The CMB with the single matter density.* Running the Boltzmann solver (CLASS) at the $\omega_m$-direct matter density $\Omega_{\rm m}^{\rm cosm} = 0.308881$ places the first three acoustic peaks at $\ell = 220$, 536, 813 (ΛCDM: 220, 536, 813), an RMS deviation of 0.18% from ΛCDM; CAMB returns the same peaks (Paper 3). The density is not fitted to the CMB: it follows from the $\omega_m$-direct identity ($\omega_c = \mathrm{KAL}_0\, \omega_b\, n_s$, fixed by $\varphi, \pi$; §VII). An earlier version contrasted this with a run that put the equation-of-state number $1 + w_0 = 0.160050$ in the density slot; $1 + w_0$ is not a density, so that run is not a version of the model and the comparison is withdrawn.

## V. STRUCTURE GROWTH: NO TENSION ON RAW DATA

The same linear EFT predicts the growth of structure, and here we report the situation honestly, including a correction to what earlier versions of this document reported. With

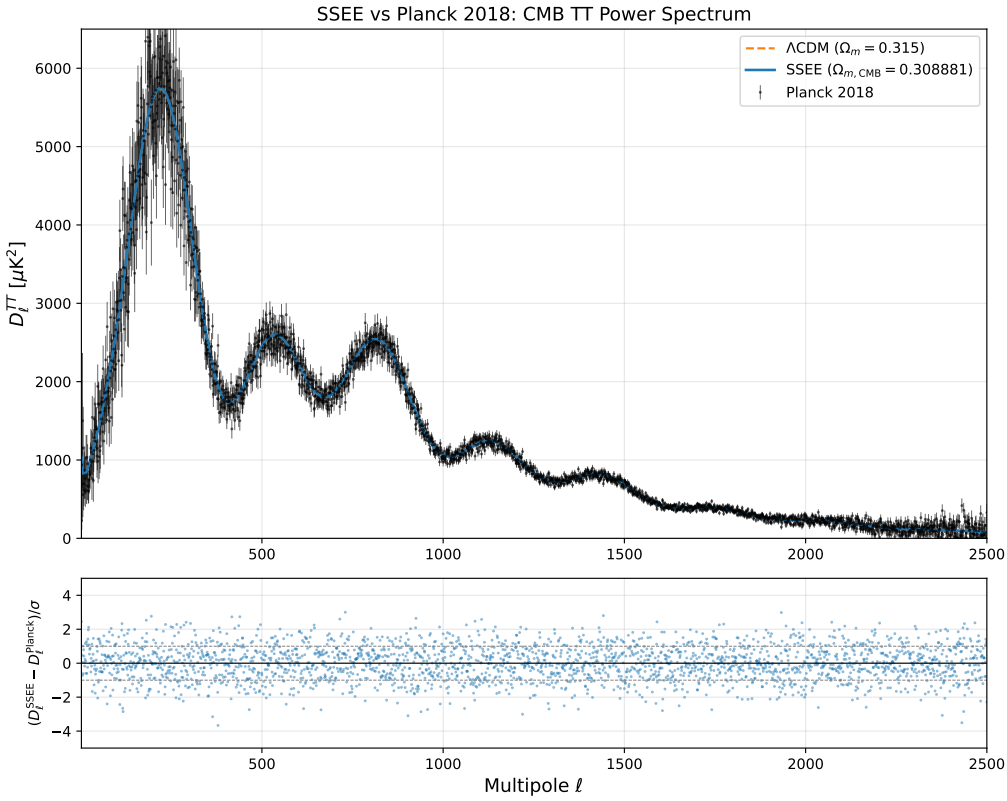



FIG. 4. CMB temperature angular power spectrum computed with `CAMB` (the Paper 3 pipeline) for the SSEE background and EFT functions (Eq. 9), compared against Planck 2018. With the dark-energy sector fixed by $\varphi$ and $\pi$, SSEE reproduces the acoustic peaks at $\ell = 220, 536, 813$ at a quality indistinguishable from ΛCDM (Table V).

the Poisson source evaluated at the CMB-calibrated density $\Omega_{\mathrm{m}}^{\mathrm{cosm}} = 0.308881$ ($\omega_m$-direct, the model's single matter density), the growth factor is essentially that of ΛCDM, $G \equiv D_1^{\mathrm{SSEE}}/D_1^{\Lambda\mathrm{CDM}} = 1.0032 \pm 0.005$, and the single-sector amplitude ceiling (CLASS forward) is

$$\sigma_8 = 0.8149 \pm 0.006, \qquad S_8 = 0.827 \tag{15}$$

when the primordial amplitude $A_s$ is held at the Planck-preferred value. That configuration was previously reported as a $3.5\sigma$ tension with KiDS-1000 [14, 15], and a two-sector $\varphi$-DM extension was proposed to resolve it. **Both the tension and the extension are withdrawn here**, for two independent reasons.

First, fixing $A_s$ to Planck is not a neutral choice: $A_s$ is one of the two free parameters this model retains at CMB level, so a number obtained that way reports the Planck–shear dis-

agreement rather than a property of the model. That disagreement is real and is not resolved here — asking this model's own CMB fit to set $A_s$, rather than borrowing Planck's, still leaves $3.5\sigma$ against the shear determination. Second, the target itself was model-conditioned: the published $S_8$ is a compressed statistic whose data-reduction pipeline assumes a ΛCDM background, so fitting to it compares a non-ΛCDM model against a partly-ΛCDM number.

Measured instead against the 225 *raw* KiDS-1000 $\xi_\pm$ data points, with the background fixed entirely by the algebra, a *single* unpartitioned matter sector $\Omega_m = 0.308881$, and only $A_s$ and the eight official nuisance parameters free, a converged MCMC ($R-1 = 0.019$, 42711 effective samples) gives

$$\sigma_8 = 0.7449 \pm 0.0186, \qquad S_8 = 0.7559 \pm 0.0189, \tag{16}$$

i.e. $0.10\sigma$ from the KiDS-1000 value $0.759 \pm 0.024$ — and, in the like-for-like comparison, $\Delta S_8 = 0.0012$ from the ΛCDM control run through the same pipeline ($0.7571 \pm 0.0194$), with $\chi^2_{\rm min} = 265.44$ over 216 degrees of freedom. **There is no $S_8$ tension to resolve, and therefore nothing for a second sector to do.**

*a. The same test against KiDS-Legacy.* The measurement above uses KiDS-1000. Repeating it against the KiDS-Legacy release—357 raw $\xi_\pm$ points, Wright et al. (2025) (arXiv:2503.19441), published sixteen months before the present analysis, over which we claim no temporal priority—closes the sector. With the same algebraic background and $A_s$ free, the shear asks for $\log(10^{10}A_s) = 3.0255 \pm 0.0396$, against the 3.04483 that the CMB fixes *under that same background*: a distance of $0.49\sigma$. The control with the Planck background held equally rigid gives $1.01\sigma$. Fixing $A_s$ to the CMB value therefore costs nothing:

$$\chi^2 = 417.97 \;\; (8 \text{ free, all nuisance, } \textit{none} \text{ cosmological}) \quad \text{vs.} \quad 418.34 \;\; (A_s \text{ free}), \tag{17}$$

over the same 357 points, giving $\Delta\mathrm{BIC} = +6.24$ in favour of the fixed-amplitude model, and +6.86 against the ΛCDM-background control (418.95). With no free cosmological parameter left, $S_8 = 0.8273$ is a prediction rather than a fit, against $0.8265 \pm 0.0176$ measured—$0.04\sigma$.

Two cautions belong with this result. The three $\chi^2$ values span 1.0 over 357 points: **the shear does not discriminate between these models**, and we do not claim that it does. What separates them is the number of parameters each had to spend and whether $A_s$ stays consistent between the CMB and the growth epoch. And the earlier KiDS-1000 release asked

for $\log(10^{10}A_s) = 2.864 \pm 0.050$, $3.6\sigma$ from the same target; the two releases differ from each other by $2.5\sigma$. What changed is the data—the $n(z)$ calibration, by the collaboration's own Appendix I—not the model. A future release returning to the lower amplitude would reopen the sector.

The $\varphi$-DM component and its particle are retracted independently of this: the subtraction that defined the second sector's density, $\Omega_{\varphi\mathrm{DM}} = \Omega_{m,\mathrm{CMB}} - \Omega_{m,\mathrm{dyn}}$, took a measured density minus $1 + w_0$ — a number from the equation of state, not a density. The subtraction is arithmetically well formed and physically empty, so the particle had nothing to be made of.

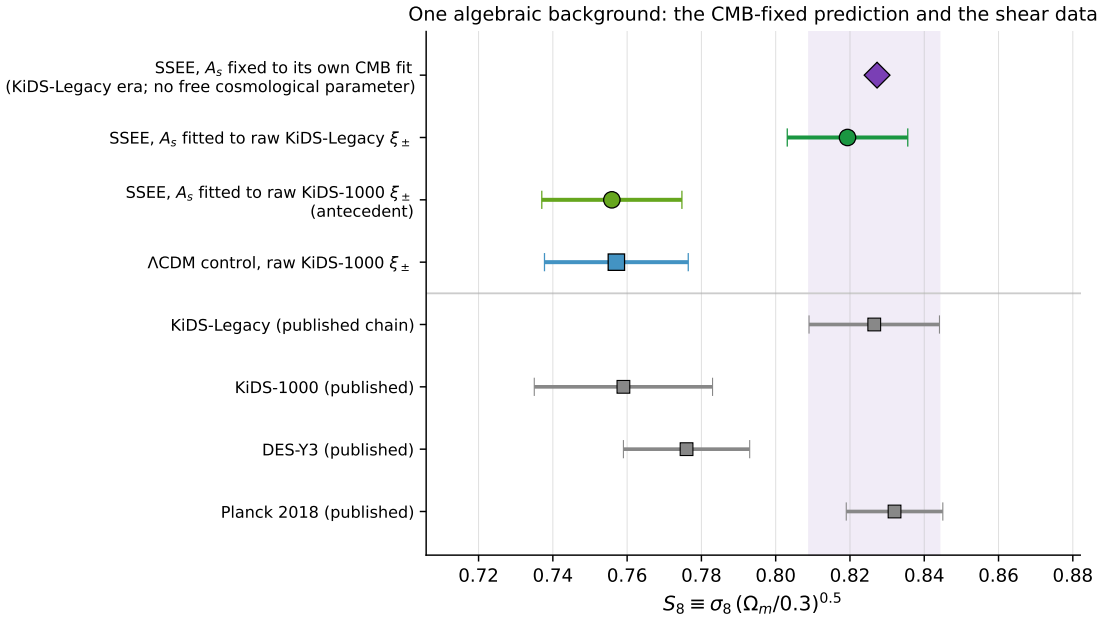



FIG. 5. $S_8$ on the algebraic background of SSEE, identical in every SSEE row; only the dataset asked to fix $A_s$ changes. Purple diamond: the prediction with $A_s$ fixed to the model's own CMB fit — no free cosmological parameter — $S_8 = 0.8273$, $0.04\sigma$ from the KiDS-Legacy posterior ($0.8265 \pm 0.0176$, purple band). Green: $A_s$ fitted to the raw KiDS-Legacy $\xi_\pm$ ($0.8193 \pm 0.0162$) and, as antecedent, to the raw KiDS-1000 $\xi_\pm$ ($0.7559 \pm 0.0189$, $0.10\sigma$ from KiDS-1000). Blue: the ΛCDM control on the same raw KiDS-1000 data ($0.7571 \pm 0.0194$). Grey: published KiDS-Legacy, KiDS-1000, DES-Y3 and Planck 2018 constraints with $1\sigma$ errors. The shift between the two KiDS releases is in the data, not in the model.

*b. The honest statement.* What resolves $S_8$ here is not additional freedom — the background is *more* constrained in this model than in ΛCDM, with $A_s$ the only quantity able to move — but the removal of two things that should not have been there: a fixed $A_s$, and a

target statistic carrying ΛCDM in its reduction. The cost of the correction is real and worth stating plainly: a pre-registered, genuinely falsifiable $P(k)$ prediction is gone, withdrawn not because data refuted it but because the construction generating it was not physical. What is gained is one postulate fewer and four open problems closed by dissolution. A full non-linear treatment still requires $N$-body simulations beyond the scope of this work.

## VI. FALSIFIABILITY AND THE PREDICTION REGISTER

The defining property of a parameter-frugal framework is that, once the algebraic invariants are fixed, the numbers it predicts are not free to move. A fit can always be improved by adding a parameter; SSEE has none to add. We therefore commit, in advance, to a register of quantitative predictions, each attached to the observation that will confirm or refute it. We separate the register into the *core* predictions that follow directly from the dimensionless invariants $\varphi$ and $\pi$, and the growth, UV and strong-gravity results that build on them; no extension sector is involved in either (the one proposed earlier is retracted, §VII).

### A. Master register: one line per observable

Before the two tiers in detail, Table VI gives the entire suite on a single page: every confronted observable, the sector it belongs to, the paper that derives it, its algebraic value, and its epistemic status (postdiction / retrodiction / prediction / measured input). The horizontal rule separates the *core* tier—fixed by $\varphi$ and $\pi$ alone—from the results that build on it (growth, UV completion, strong gravity). This is the navigation map for the whole programme; each row is expanded in the paper named in its last-but-one column and in the Predictive Register of Paper 1.

### B. Core predictions (clean from $\varphi, \pi$)

These quantities carry no fitted coefficient. They are bare ratios of the golden mean and $\pi$, and they cannot be adjusted to accommodate data.

The tensor-to-scalar ratio deserves emphasis. Because $n_s = 1-\varphi^{-7}$ and $r = \varphi^{-10}$ are postulated to be of this form, the consistency relation of the underlying $\alpha$-attractor closes alge-

| Observable | Sector | Value | Paper | Status |
|---|---|---|---|---|
| *Core tier — clean from $\varphi, \pi$ (no fitted coefficient)* | | | | |
| $w_0 = -T_r/M_v$ | Background (DE) | $-0.839950$ | P1,2,4 | Postdiction |
| $w_a = -P_{\rm sc}/I_g$ | Background (DE) | $-0.669975$ | P1,2,4 | Postdiction |
| $(w_0, w_a)$ vs DESI DR3 | Background (DE) | same fixed point | P1,2 | **Prediction (pre-reg.)** |
| $H_0^{\rm glob} = H_0^{\rm SH0ES}(1 - f_{\rm scr})$ | Background (DE) | $67.962\,{\rm km\,s^{-1}\,Mpc^{-1}}$ (target $3(\varphi+\pi)^2$) | P9,10 | Derived (SH0ES input) |
| BAO distances, background fixed | Background | $\chi^2 = 11.41/13$ ($\Lambda$CDM-Planck 31.98; Tab. IV) | P2 | Postdiction |
| $\omega_c = {\rm KAL}_0\,\omega_b\,n_s$ | CMB | 0.119514 | P1,3 | Retrodiction |
| $\Omega_{\rm m}^{\rm cosm} = \omega_m/h^2$ | CMB | 0.308881 | P1,3 | Retrodiction |
| $n_s = 1 - \varphi^{-7}$ | Inflation / CMB | 0.965558 | P1,3 | Retrodiction |
| $r = \varphi^{-10}$ | Inflation | $8.1 \times 10^{-3}$ | P1 | **Prediction** |
| $\alpha_T = \alpha_M = \alpha_B$ | EFT | 0 (exact) | P7 | Retrodiction |
| $\alpha_K = 3\Omega_{\rm DE}(5 - 3w_0)$ | EFT | 15.591336 | P7 | Derived (unobservable on linear scales) |
| $\gamma_{\rm IS}$ (Israel–Stewart) | Growth | 0.550 | P5 | Retrodiction |
| $f\sigma_8$ (BOSS DR12, raw multipoles) | Growth | $\Delta\chi^2 = -2.03$ vs. $\Lambda$CDM, equal freedom | P6 | Retrodiction |
| *Built on the core — growth, UV completion, strong gravity (no extension sector)* | | | | |
| $S_8$ (KiDS-Legacy, $A_s$ fixed by CMB) | growth | 0.8273 predicted vs. $0.8265 \pm 0.0176$ ($0.04\sigma$) | P6 | **Prediction** |
| $M = (5\varphi^8\rho_{\rm crit})^{1/4}$ | UV compl. | 9.68 meV | P10 | Derived (UV) |
| Lensing (canonical EFT) | Strong gravity | GR+DM (uncond.) | P8 | Retrodiction |
| *Measured input (not derived)* | | | | |
| $A_s$, $\tau$ | CMB amplitude | sampled ($k = 2$) | P1,3 | Measured |
| $\rho_{\rm vac}$ (abs. scale) | Vacuum | single measured input | P1,10 | Measured |

TABLE VI. Suite-wide master register: one row per observable across the ten-paper programme. "Postdiction": algebraic result fixed by the closed dictionary, then compared to already-public data (no claim of temporal priority). "Retrodiction": derived independently, then compared to an existing measurement. "Prediction": fixed now, tested against not-yet-released data. "Measured": an input, not a derivation. "Internal check": a same-paper consistency quantity, not itself tested against external data. No optimisation over the algebraic formula space was performed. The detailed per-sector registers follow below and in the Predictive Register of Paper 1.

braically: with $N_* = 2\varphi^7$ and $\alpha = \varphi^4/3$ one obtains $r = 12\alpha/N_*^2 = 12(\varphi^4/3)/(2\varphi^7)^2 = \varphi^{-10}$ identically. The prediction is thus internally locked rather than independently adjustable. Its value, $r \simeq 8.1 \times 10^{-3}$, lies squarely within the reach of LiteBIRD and CMB-S4, which target $\sigma(r) \sim 10^{-3}$. SSEE will be decisively tested by the next generation of $B$-mode experiments.

*a. A pre-registered DESI DR3 test.* The equation-of-state point is fixed forever at $(-0.840, -0.670)$; each DESI release is therefore a fresh blind test of the *same* target. The trajectory to date: against the legacy DR1 posterior the point sat at $\sim 0.05\sigma$; against DR2,

| Observable | SSEE value | Algebraic form | Falsifying probe |
|---|---|---|---|
| $w_0$ | $-0.839950$ | $-T_r/M_v$ | DESI Y3 / Euclid BAO+SNe |
| $w_a$ | $-0.669975$ | $-P_{\rm sc}/I_g$ | DESI Y3 / Euclid BAO+SNe |
| $n_s$ | 0.965558 | $1-\varphi^{-7}$ | Planck / CMB-S4 |
| $r$ | $8.1306\times10^{-3}$ | $\varphi^{-10}$ | LiteBIRD / CMB-S4 |
| $c_s^2$ | 0.021284 | $(M_v-T_r)/(5M_v+3T_r)$ | DESI / Euclid growth |
| $\alpha_T,\alpha_M,\alpha_B$ | 0 | exactly zero | GW standard sirens, ISW |

TABLE VII. Core prediction register. Each value is fixed by $\varphi$ and $\pi$ alone, with no free parameter available to absorb a discrepancy. The tensor ratio $r=\varphi^{-10}$ sits an order of magnitude below the current bound $r<0.036$ and is the cleanest single falsification target: a detection above $\sim10^{-2}$, or a robust upper bound below $\sim8\times10^{-3}$, refutes the inflationary sector.

whose uncertainties are ∼40% tighter, it sits at $0.24\sigma$ (Pantheon+; §II B)—the contour shrank and the fixed point stayed inside the 68% region, which a fitted model can always do but a rigid one need not. We pre-commit, in advance of the DR3 release (expected 2027), to the following quantitative expectations for DESI DR3, whose uncertainties are forecast to improve by ∼1.5×: if the DR2 central values persist, the point will sit at $\sim0.5\sigma$ of the DR3+CMB+Pantheon+-class posterior (still inside $1\sigma$), while the Union3/DESY5-class combinations would sharpen to $\sim2.2$–$2.7\sigma$ —DR3 will therefore *discriminate among the supernova compilations* whose spread already dominates the range quoted in §II B. The falsification line is pre-registered *now*, before the DR3 release, and fixed to a specific supernova compilation so that no post-hoc choice of sample can rescue the model: a joint exclusion of $(-0.840,-0.670)$ at $>3\sigma$ in the DR3 $w_0$–$w_a$ plane for the DR3 + CMB + *Pantheon+* combination —the same Pantheon+ compilation used for the headline agreement throughout this work—refutes the algebraic background. We will additionally report the Union3- and DESY5-class combinations for completeness, but the pre-committed falsifier is the Pantheon+ line named here. No parameter exists with which to absorb such an outcome.

The vanishing of the EFT slip functions $\alpha_T=\alpha_M=\alpha_B=0$ is equally sharp: SSEE predicts a gravitational-wave speed identical to light ($c_T=c$) to all orders in the dark sector, no running of the effective Planck mass, and no braiding. Any measured departure—an anomalous standard-siren distance, a time-varying $G_{\rm eff}$, or a non-zero braiding signature

in the ISW–galaxy cross-correlation—falsifies the canonical EFT presented in §IV.

## VII. HONEST ACCOUNTING AND OUTSTANDING CHALLENGES

A framework is only as credible as its honesty about what it does not yet explain. We therefore state plainly the inputs the model still requires, the exact parameter count, and the principal open problems—not as caveats buried in a footnote, but as the research agenda that the present work both motivates and constrains.

### A. What is measured, not derived: the vacuum scale

SSEE does not predict the absolute energy density of the dark sector. The overall scale of the vacuum, $\rho_\Lambda \simeq 6 \times 10^{-10}\ \mathrm{J\,m^{-3}}$, enters as a single measured input, which we label *Postulate D*. What the algebra fixes is the *shape* of the dark-energy sector—its equation-of-state ratios $w_0, w_a$, the matter density $\omega_m$, and the EFT function profile—not its amplitude. We make no claim to have solved the cosmological-constant problem; we have moved its free numbers from six tuned cosmological parameters down to a single measured normalisation plus the dimensionless invariants $\varphi$ and $\pi$. This is a reduction in arbitrary inputs, not their elimination.

### B. Parameter count: $k = 2$ versus 6

We claim a *minimal-parameter* framework, never a zero-parameter one. The honest accounting is the following. ΛCDM carries six fitted parameters $(\Omega_b h^2, \Omega_c h^2, \theta_*, \tau, n_s, A_s)$. SSEE fixes four of them algebraically from $\varphi$ and $\pi$: the baryon density $\omega_b$ (OP-1), the CDM density via the forward identity $\omega_c = \mathrm{KAL}_0\, \omega_b\, n_s$, the spectral index $n_s = 1 - \varphi^{-7}$, and the expansion anchor $H_0 = 67.962\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ (whose value in these units coincides with $3(\varphi+\pi)^2$; derived, Paper 9). This leaves exactly two free CMB parameters—the primordial amplitude $A_s$ and the optical depth $\tau$—a $k = 2$ versus $k = 6$ comparison, which is the count used in the Paper 3 Bayesian model selection. The dark-energy ratios $(w_0, w_a)$ and the EFT functions are predicted, not fitted. On the dark-sector extension there is nothing left to promise: the $\varphi$-DM mass coefficient (OP-9) and the non-minimal coupling $\xi$ (OP-11) are

retracted with the sector itself, and both close by dissolution, tracked separately as open problems rather than counted among the fit parameters.

### C. Open problems as bridges to future work

The open problems that bear on this work are listed below; one of them has closed since earlier versions and is kept for the record. They define the path from the present minimal-parameter framework toward a fully derived theory.

- **The CMB matter density's algebraic origin (OP-1).** With the former MIRA matter bridge *dissolved* (the $\omega_m$-direct reframe, §II D; OP-8 retired), the CMB matter fraction $\Omega_{m,\mathrm{CMB}} = 0.308881$ now rests directly on two algebraic inputs: the baryon density $\omega_b = (\pi - \varphi)/[3(\varphi + \pi)^2] = 0.022418$ and the forward identity $\omega_c = \mathrm{KAL}_0\, \omega_b\, n_s = 0.119514$ ($0.40\sigma$ from Planck). With it the CLASS run of §IV C reproduces the acoustic peaks of ΛCDM. What remains open is a first-principles derivation of these algebraic forms from the field action; the burden has moved from "derive MIRA" to "ground $\omega_b$ and the $\omega_c$ identity." The values are not fitted, but their UV origin is open.

- **The $\varphi$-dark-matter mass (OP-9): closed by dissolution.** This entry previously recorded an algebraic mass $m_\varphi$ whose UV origin was open. The particle itself is now retracted, so there is no multiplier left to derive and the open problem closes without having been solved. The free-streaming step in $P(k)$ that descended from it is withdrawn as a prediction. OP-10, OP-11 and OP-12 close the same way.

The growth-of-structure status of §V no longer mirrors this structure. The $S_8$ "tension" at 0.827 was an artefact of holding $A_s$ at the Planck value while comparing against a ΛCDM-conditioned summary statistic; fitted to raw shear data with a single sector it is $0.7559 \pm 0.0189$, matching the ΛCDM control on the same data ($0.7571 \pm 0.0194$) with four fewer free parameters. Against the later KiDS-Legacy release the growth sector needs no free amplitude either: the $A_s$ the shear asks for sits $0.49\sigma$ from the one the CMB fixes under the same background, and fixing it predicts $S_8 = 0.8273$ against $0.8265 \pm 0.0176$ measured. No extension sector is required, and the one proposed is retracted. The non-linear treatment required to settle the matter definitively still demands $N$-body simulations beyond this work's scope.

None of these gaps touch the algebraic core of Table VII: the dark-energy equation of state, the spectral index, the tensor ratio, and the vanishing EFT slip functions stand on the dimensionless invariants alone and are falsifiable today.

## VIII. CONCLUSIONS

We have presented a minimal-parameter framework in which the shape of the dark sector is fixed by the two dimensionless constants $\varphi$ and $\pi$. The late-time equation of state emerges as bare ratios of structural invariants, $w_0 = -0.839950$ and $w_a = -0.669975$, and matches the DESI DR2 posterior at $0.24\sigma$ (Pantheon+) without a single fitted dark-sector parameter. The effective-field-theory description is canonical, with vanishing tensor, running, and braiding functions, $\alpha_K = 15.591336$ and $c_s^2 = 0.021284$; the inflationary sector predicts $n_s = 0.965558$ and $r = 8.1 \times 10^{-3}$; and a Boltzmann calculation with its single matter density places the acoustic peaks where ΛCDM does.

The framework's strength is its frugality and its consequent falsifiability. Its core predictions—the dark-energy ratios, the spectral index, the tensor ratio, and the vanishing slip functions—rest on $\varphi$ and $\pi$ alone and will be tested decisively by DESI Y3, Euclid, LiteBIRD, and CMB-S4 within this decade. We have been equally explicit about its limits: the vacuum scale is a measured input and the algebraic origin of the CMB matter density ($\omega_b$, $\omega_c$) is unresolved. The growth amplitude, once reported here as a tension with weak lensing, is not one on raw shear data: with $A_s$ fixed to the model's own CMB fit, the growth sector predicts $S_8 = 0.8273$ against KiDS-Legacy's $0.8265 \pm 0.0176$. These define a concrete research agenda rather than fatal flaws, and none of them touch the algebraic core.

Whether the deep agreement between a handful of golden-ratio combinations and the measured dark sector reflects an underlying structural principle or a remarkable set of coincidences is precisely the question that the prediction register of §VI is designed to settle. The next generation of cosmological surveys will provide the answer.

### Appendix A: The two lineage laws of the constant dictionary

The look-elsewhere test of §II C rests on the dictionary being a *closed, finite* set fixed in advance, rather than an open search space. This appendix states the generative rules

that make it so. We present them as a descriptive grammar of the Genesis set [1]: they reproduce the dictionary exactly, but their *dynamical* origin is not claimed and remains the open mechanism of §VII. Stating them explicitly is what turns "55 named invariants" from an assertion into a checkable construction.

*a. Seeds and scaffold.* Everything descends from two dimensionless seeds, $\varphi = (1+\sqrt{5})/2$ (UV) and $\pi$ (IR). Their sum is the scaffold $\Omega = \varphi + \pi = 4.759627$, and their mean is the bifurcation point $\beta \equiv \mathrm{BIAL} = (\varphi+\pi)/2 = 2.379813$ at which the dictionary splits into two branches.

*b. Law I — the copy law ($\varphi$-branch).* The $\varphi$-branch is generated by a single root, the screening amplitude $\mathrm{AURA} = \beta + \varphi = 3.997847$, which propagates *only* by integer multiples and the dyadic half. These are the dimensional ceilings:

$$\begin{aligned} \mathrm{MIRA} = \tfrac{1}{2}\mathrm{AURA}, \quad \mathrm{AURA}, \quad \mathrm{DUAL} = 2\,\mathrm{AURA}, \\ \mathrm{TRIAL} = 3\,\mathrm{AURA} = T_r, \quad \mathrm{CUARTAL} = 4\,\mathrm{AURA}. \end{aligned} \tag{A1}$$

The denominator of $w_0$ is $M_v = \varphi + \pi + K_v$, a structural convergence built by lineage on $K_v\,(\mathrm{KRYSTOS} = \varphi + \pi + \Omega)$; it *coincides in value* with $3\Omega$ but is not the scaffold tripled—the construction, not the magnitude, is what fixes its identity. With the copy-law wall $T_r = \mathrm{TRIAL} = 3\,\mathrm{AURA}$ as numerator,

$$|w_0| = \frac{T_r}{M_v} = \frac{\mathrm{TRIAL}}{\varphi + \pi + \mathrm{KRYSTOS}} = 0.839950, \tag{A2}$$

which reduces *numerically* to $\mathrm{AURA}/\Omega$—an identity of value between distinct named entities, not a construction by scaffold multiples. So $w_0$ and $w_a$ are ratios internal to the closed dictionary, not two independent draws.

*c. Law II — the no-self-sum law ($\pi$-branch).* The $\pi$-branch is generated additively: each child is a sum of previously defined parents, subject to the constraint that *a node never adds to itself*. Thus $\mathrm{KAL} = \beta + \pi$, $P_{\mathrm{sc}}\,(\mathrm{PYROS}) = \Omega + \varphi$, $\mathrm{IGNIS} = \pi + P_{\mathrm{sc}}$, and so on, building up to the convergences. A corollary is that two distinct lineages may land on the *same value* without being the same entity: $\mathrm{IGNIS} = \pi + \mathrm{PYROS}$ (disruptive, $\pi$-branch) and $K_v\,(\mathrm{KRYSTOS}) = \varphi + \pi + \Omega$ (structural) both equal 9.51925, yet are distinct nodes; likewise $\mathrm{SOLAR} = \mathrm{MAR} = 7.90122$. The look-elsewhere count of §II C treats these honestly, deduplicating by value so the 55 named nodes collapse to 25 distinct numbers. The second equation-of-state ratio lives entirely on this branch, with the denominator the $\pi$-branch node

IGNIS (not the structural $K_v$, though the two coincide in value),

$$|w_a| = \frac{\text{PYROS}}{\text{IGNIS}} = \frac{P_{\text{sc}}}{\pi + P_{\text{sc}}} = 0.669975. \tag{A3}$$

*d. Closure.* The set is finite because both laws terminate at the four-dimensional integration invariant $M_v = \varphi + \pi + K_v = 14.278880$ (a Sovereign convergence coinciding in value with $3\Omega$), the maximal dimensional invariant; no node exceeds it. This termination—not a choice made to fit data—is what bounds the dictionary to a fixed, enumerable set and licenses the conservative look-elsewhere denominator of §II C. Any future extension would proceed only by the same copy law (the planned dimensional copies QUINTAL–DECAL), which, as noted in Table II, add no accidental hit within the observational band of either $w_0$ or $w_a$.


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## DATA AND CODE AVAILABILITY

The SSEE framework is fully transparent and reproducible. The complete code repository, including the CLASS and CAMB configuration files, the COBAYA MCMC chains, the explicit record of Open Theoretical Problems (OP), and the cosmological validation ledgers, are publicly available at the GitHub repository https://github.com/mikealmeida1721/SSEE-Cosmologia and permanently archived on Zenodo, DOI 10.5281/zenodo.19654118 (concept DOI, all versions).

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